3- Tier architecture (Fronted, backed, middleware)

2- Tier architecture (Fronted, backed or middleware)

n- Tier architecture.

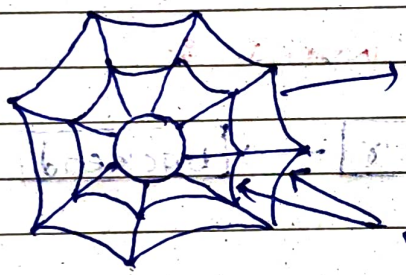
- Fronted :-

HTML, CSS, Javascript, Bootstrap, React } Web Technology ← users

a. Webpage
b. website
c. Web

2010 Web application

- Web :- Interconnection of data.



- Fronted ⟷ middleware ⟷ Backend
  - middleware ↓ Java / python
  - Backend ↓ SQL

- Web terminologies :-

1) Web → It is interconnection of data where we can store as well as access the data.

2) server :- It is S/W / H/W, which receives request from user sends the request to database through programming.

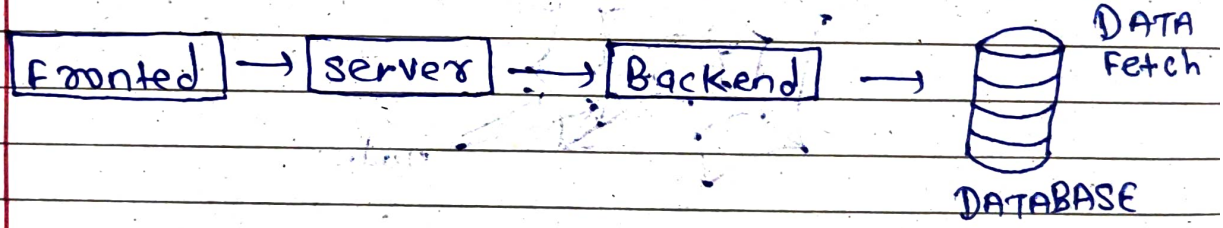
logic & fetch the data and respond it to user.

3] Browser :- It is standalone application we run frontend code instead of using third party application use own platform such as various browsers.

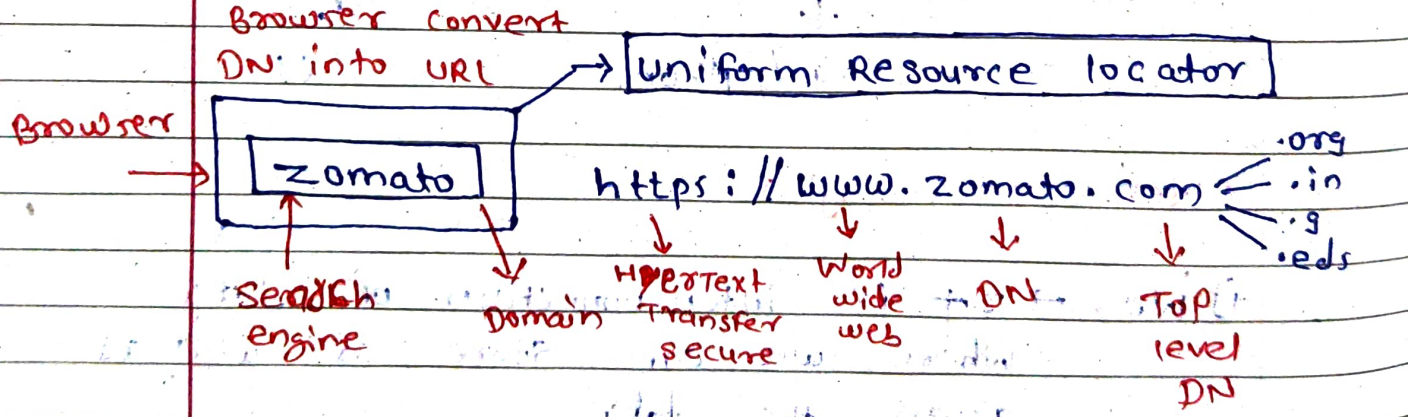
4] Webpage :- A digital document which consist information such as text, image, media, google, maps, hypertext, icons.

5) Websites :- It is collection of Webpages.

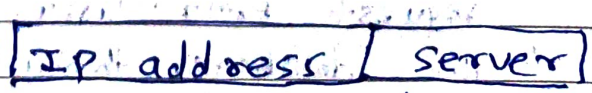
* How Servers Works :-



* How Browser Works implicity. (Web application)



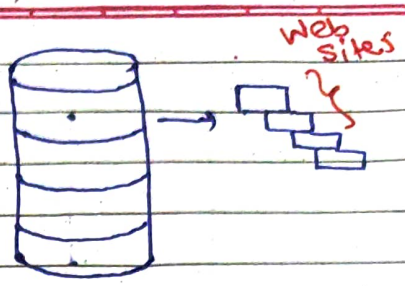
Domain name system → DNS → Convert URL into IP Address



1.00.23.55.78 / 5500

1] If IP address matches
→ OP

2] If IP address not matches
→ 404 page not found.



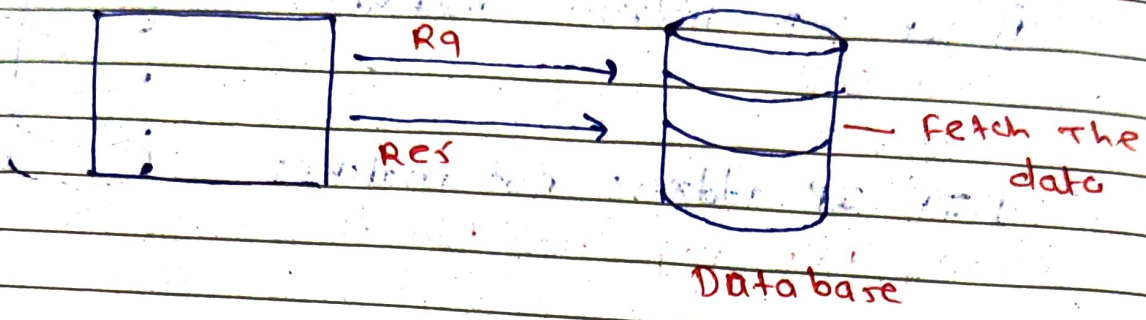
3] Issue → spending / loading

Two Type of Websites.

User friendly	User Interactive
Stylish Styling eg. Instagram Snapchat	logic Building eg. 1] government / Web 2] zomato 3] Flipkart.

* Types of websites.

Static Website	Dynamic Websites.
In static Websites user can't modify data.	In Dynamic Web sites user can change or modify data.
Data will be same for all users	Data will be diffent for all users.
Reloading time is less	Reloading time is more
only specific component or Page is reloaded	entire website will be reloaded.
W3 schools, educational Web sites	e.g :- Instagram.



* Fronted structure :-

Web dev
- HTML → structure
- CSS → styling
  - Tailwind CSS
  - bootstap
  - mataial ui
- JS → functionality
  vote.JS

Imp UI library

React.JS
- next.JS
- next.JS
- next.JS

Angular.JS

M → mongoDB → DATAbase

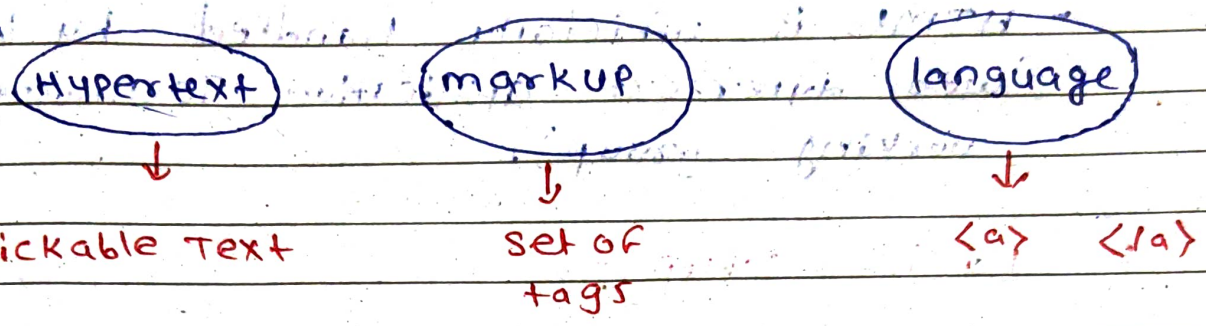
E → Express → Framework for node.JS

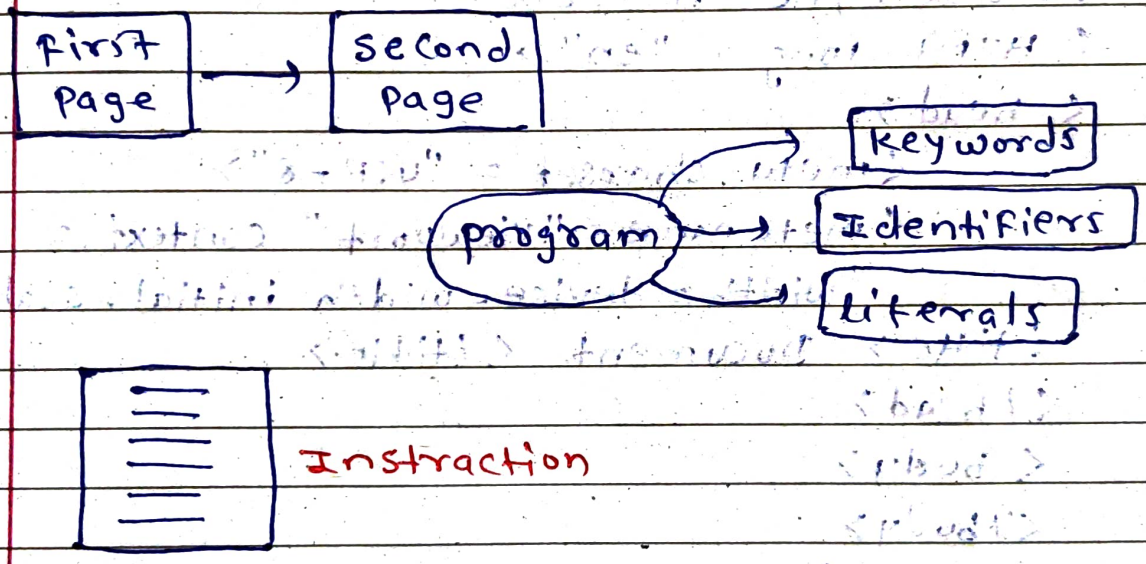
A/R → React

N → Node.JS → middleware

- HTML :-

HTML is a Hypertext markup language and also root frontend language reference link to usualize the data on UI (user Interface) with the help of some predefined tags which are browser understandable.





- History of HTML :-

-HTML language was originally generated by XML (reference)
The rounder of w3c (world wide web consoutiun)
Tim Berners lie the got the idea in 1989.

Then He introduced versions as follows

HTML 1 → 1991
HTML 2 → 1993
HTML 3 → 1995
HTML 4 → 1997
HTML 5 → 2014 (latest version)

- HTML is officially handled by WHATWG (web Hypertext application Technology working group).

* Basic setup :-

```
<!DOCTYPE HTML>
<HTML long = "en">
<head>
    <meta charset = "UTF-8">
    <meta name = "viewport" context =
    "width = device-width initial-scale = 1.0">
<title> Document </title>
</head>
<body>
</body>
</HTML>
```

- It is declaration of which version of HTML is being used. (HTML 5)

<HTML>
- Root tag of HTML which holds long attribute.

<head>
- Head tag is used to code the meta,

data in the from of meta tag.

- The first meta tag stands for characteristic of UTF-8 (Unicode Transformation format).
- The second metatag stands for device measuremate.

<title>
The main heading of the webpage is declared in Title tag.

* Tags in HTML :-
Tag is a predefined them or a word which is enclosed between angular brases (< >).

- There are two types of Tags.
  1] paired Tag.
  2] unpaired Tag.

1] paired tag :-

paired tag has two set of same Tags The first tag is called as opening tags & second tag is called closing tag.

2] unpaired tag :-

unpaired tags are single tag hence, it is called as self-closing tag.

* Basic Tags of HTML :-

<p> tag - p tag is paragraph tag we can use this tag To describe paragraph content on us.

<heading> tag - To deline heading Type of content we use heading tags. There are size heading Tags.

h1, h2, h3, h4, h5, H6
- All heading Tags are paired tags.

 Tag - To break. The line we use Thir br tag and it is unpaired tag

<hr> tag - To add Horizontal line after The specific line (if needed) we use Hr tag.

 tag - It is inline container & Inline element

<Div> tag - It is block container & block element in nature.

Size of heading tags.

h1 → em
h2 → 1.8 em
h3 → 1.5 em.
h4 → 1 em ⟶ Default size of content
h5 → 0.8 em
h6 → 0.5 em

16 Pixels ⇒ 1 em

Inline	Block
Inline element start in same line	Block element always start in new line & next line
Inline element occupie only content width	Block element occupies The entire width of view port
Inline element doesn't Takes height & width	Block element accept height and width.
Inline element will not accept Top & bottom margin.	Block element accept Top & bottom margin.

- **HTML ELEMENTS :**
The content enclosed between two paired tags is called HTML element.

- All unpair Tag are void element

There are 3 Types of HTML element.

1] Block element
2] Inline element
3] void element.

ex:
 content
↓
content
↓
(HTML element) All time say
 void element & unpaired tag.

* pre tag :- It is performatted text tag which define the exact code or output how we have descripted in the editor to print patten and diagrams.

* Attributes of HTML.

Attributes are the added extension to tag which provide extra information To tags.

Attributes always written or declared inside the opening tag of HTML.

* Attribute are declared in Two ways

① only Attribute
2) Attribute with value.

* Three are Two Types of Attributes.
① Global or core attributes
② elements specific Attributes.

① ① Global or core attributes
These attributes can be used in every Tag of HTML. They will have access through but The all HTML tags.

1) class
2) id
3) Style : to give style to html element
4) title : to create a tooltip to HTML element.

3) element specific Attributes :-
These attributes can be used in only specific tags tags such as img anchor and media.

1) src and Alt ⟶ img tag
2) href ⟶ anchor Tag
3) src and controls ⟶ media Tag.

* formatting tags in HTML

1) b tag :- it is blod tag and it in used to make content bold in nature.

2) Strong Tag :- it is strong Tag and it is used to make content bold as well as important for the developer.

3) i tag :- it is itallic tag and it is used to makei content italic is nature.

4) em tag :- it is emphesized tag and it is alternative for i tag and provides

5) u tag :- it is underline tag and it is used to add the underline to the content

6) ins tag :- it is insent tag and it is alternative for u tag and provides same feature an u tag.

7) Superscipt tag :- Sup tag it is used to shift the content above with it's respentive content.

8) Subscript tag :- Sub tag is used to shift The content below with it's respective content.

9) Q Tag :- it is Quatation tag which add suitable Quatation to content

10) Mark Tag :- it is used to highlight The content (default color is yellow).

11) Strike tag :- It is used to add the line through The content and it is depricated in nature.

12) center Tag :- It is used to make content in The center and it is also depricated tag.

13) marquee tag :- It is also depricated tag but still in use which create dynamic sliding effect to The HTML content.

* Attributes of marquee Tag

behavior = " " : Alternate

direction = " " : right, left, up, down.

Scrollment = " " : in pixels to increase the sliding effect.

List creation in HTML

any element or contenent we need to display one by one we make use of list inside HTML

There are four Type of list

1] ordered list
2] unordered list
3) description list
4) nested list

1) ordered list :- in ordered list we print or display the items in sequence or in order such as number's, alphabets and roman number.

OL TAg :- It is used to create root Tag.
LI tag :- It is used to create list items.

Ex:-
```
<ol Type = "A" start = "520" reversed>
    <li> Kaju katli </li>
    <li> vadapav </li>
    <li> samosa </li>
    <li> Pav bhaji </li>
</ol>
```

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Attributes of ordered list.

1. Type = " " : here we mention what Type of order we need.

2. Start = " " : here we need to mentation number value which indicates the stating positive of the list

3. reversed : This Attributes make your list reversed.

2. Unordered list : in unordered list we print or display the items one after one with the helps of some markers such as disc, square, circle and none

UL Tag : it is used to create root tag of unordered list

LI tag : it is used to create list items.

for ex.

```
<UI Type = "circle">
    <li> Speed 400 </li>
    <li> Ducati </li>
    <li> Jawa </li>
</ul>
```

Attributes of ordered list.

1. Type = " " : here we mention what Type of order we need for ex; disc, square, circle and none.

3. description list : here we create description list by using dl, dt and dd tag.

dl tag : To create description list root Tag.
dt Tag : To create description term Tag
dd tag : To create desciption data tag

4) Nested list : A list inside another list is called as nested list.

example of description list and neored list Together.

```
<dl style = "margin-left : 100px;" >
    <dt> Husband Qulities </dt>
    <dd> Good Job </dd>
        <ul style = "margin-left : 50px; ">
            <li> Govt Job </li>
            <li> High salary </li>
            <li> Work which you do honestly </li>
        </ul>
    <dd> cooking is must </dd>
    <dd> understanding </dd>
    <dd> loyal </dd>
    <dt> Wife Qualities </dt>
    <dd> Ladki and zinda </dd>
</dl>
```

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* Table creation in HTML

Table is logical structure of rows, column and cell which is used to display data in systematic manner.

Table Tag: it is root Tag of Table creation.

tr Tag :- It is used to create table row

th Tag :- It is used to Table heading (by default it appers in bold and center of the cell.

td tag :- It is used to define Table data.

- Attribute creation in HTML. TABLE.

border = "": it is used to insert border To entire Table the values given in numbers or pixels.

cellpadding = "": with the help of this Attribute we can increase the space inside the cell, the values given in number or pixels.

cellspacing = "": With the help of this Attribute we can increase the space bet" Two or more cells; the values given in numbers or pixels.

specific tag Attributes

rowspan = "" : with the help of this Attribute we can merge (expand) two or more rows inside column.

colspan = "" : with the help of this attribute we can merge (expand) two or more column inside row.

* Image creation :→

Images are the very crucial part of displaying the information in easy way. To print Images we use.

img Tag : It is unpaired tag and inline element in nature

Attribute of img tag

src = "" : src means source, here the path or address or source will be given (absolute or relative both.).

alt = "" : it is alternative name for img, if image is unable to reload then alternative name appears.

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* Anchor tag in HTML :
Anchor Tag in HTML is paired tag which converts hypetext into hyperlink which redirects one page to another page.

plain text → normal text.
Hypertext text → clickable text.

3 default color of anchor tag.

1. Unvisited color : purple
2. active color : red
3. visited color : blue.

anchor tag Attributes.

href = " " : it is hyperrefrence Attribute where path or url will be given as value.

target = " " : This attributes indicates the redirection The various pages either on same page or different page.

values are :
- _blank
- _self
- _parent
- _Top

Advance features of anchor Tag.

download Attribute : if we provide this attribute The specific file will be downloaded on displayed on UI.

mailto : This feature allow us to mail to the person whose mail is given in href by end user

media tags :-

media tags :- media tags are used to insert different type of media in our webpage such as audio video and iframe.

audio tag :- This tag is being used to add or insert audio inside webpage.

Attributes of audio tag :-

① src :- This attributes will be present in audio tag where we can add only relative path or saved address in it.

2) controls :- provides control mechanism bar to control audio.

3) autoplay :- by adding this attribute we can start play audio automatically.

4) muted :- by default only the audio will be in mute state.

5) loop :- if we add this attribute the specific audio will play in loop (again and again).

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* video tag :- this tag is being used to add or insert video inside webpage.

Attributes of video tag :

1) src (source) :- This attribute will be present in video tag where we can add only relative path or saved address in it

2) controls :- provide controls mechanism bar to control video.

3) autoplay :- by adding this attribute we can start play video automatically.

4) muted :- by default only the video will be in mute state

5) loop :- if we add this attribute the specific video will play in loop (again and again).

* iframe tag :- This tag is used to insert some another document or own document inside our own document or webpage which can be with absolute path as well as relative path.

Ex:- youtube videos, google map

Semantic Tags :- (Elements).

Sematic tags are the one who provides or explicitly tells the purpose of using the tags.
because using div and span tag will not allow us exact meaning of it by it's name.

There are few semantic tags such as:

① The <article> Tag :- The article Tag is used for content that stands alone and can be independently distributed or reused such as a blog post or news article.

② The <aside> Tag :- is used to place content in a sidebar i.e aside from the existing content it is related to surrounding content.

3) The Details and Summary tag.
the details defines additional details that the user can hide or view "summary" defines a visible heading for a details element.

4) The header Tag.
As the name suggests it is a for the header of a section introductory of a page There can be multiple headers on a page.

5) The footer Tag:-
footer located at the bottom of any article or document, they can contain Contact details, copyright information etc. There can be multiple footers on a page.

6) The main Tag:-
It defines the main content of the document the content inside the main tag should be unique.

7) The Table Tag:-
This Tag is used to define or create the Table structure of html.

8) The section Tag.
A page can be split into section like introduction contact information Details etc and each of these section can be in a different section Tag.

9) The nav Tag:-
The nav Tag is used to define a set of navigative links in the form of navigation bar or nav menu.

10) form Tag:-
This Tag is used to define or create form to accept user inputs.

To remember all semantic tags H-DAMN-STAFF - header, details, aside, main, nav, section, Table, article footer, form.

* FORM CREATION IN HTML.

form tag is very important tag of HTML where it accepts the user input from the user itself which we can send it to backed or server for next process.

It is mainly used for!
Login forms
Registration forms
contact forms
search forms
payment forms.

basic tags of form

form tag :- it is root tag of form and also paired tag.

1) input tag :- it is unpaired tag. the <input> tag is used to create different types of input fields.

2) label tag :- the <label> tag defines a label for an input element.

for ex:- <label for = "user" > username : </label>
<input type = "text" id = "user" >

3) textarea tag :- the <Textarea> Tag is used to create a multi-line text input field. Suitable for address, comments and feedback.

4) select Tag :- The <select> Tag creates a dropdown list. Also it is paired tag and root Tag for dropdown.

5) option tag :- The <option> Tag defines an different options inside a dropdown list

6) button tag :- The <button> tag creates a clickable button.

7) Fieldset Tag :- The <Fieldset> Tag Groups related form elements Together, kind of border to entire form.

8) legend tag :- The <legend> tag defines a caption or heading for a <Fieldset>

9) optgroup tag :- This makes an groups of dropdown options.

BASIC ATTRIBUTES of FORM TAG :-

for and id attribute :- for and id attribute are with value attribute and to interconnect label tag with respective input tag we use for and id attribute.

name attribute :- The name attribute value is acts like key for the input data (key value pair data) and shows the data in navigator for developer.

value attribute :- This attribute holds the actual value for respective radio, checkbox and dropdown types. specifically whenever we have multiple option to select

Type = "" : In this attribute we mention the specific datatype of input data such as text, number, email, date, datetime, datetimelocal, password, radio, checkbox, file, range, color reset submit, mouth.

VALIDATION ATTRIBUTES of FORM.

placeholder :- This attributes holds the value which appears as false text or faint text in input box.

required :- makes a field mandatory, the form cannot be submitted if this field is empty. and it is without value attribute

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minlength :- sets the minimum number of characters allowed.

maxlength :- sets the maximum number of characters allowed.

max : sets the maximum value for number or date inputs.

min :- sets the minimum value for number or date inputs.

pattern :- Defines a regular expression pattern that must match syntax pattern = "[A-Za-z0-9@#$] {length in numbers}"

readonly :- makes field non editable. user can see the value but cannot change it. and it is without value attribute.

disabled :- user can see the value but cannot change it. Disables the field completely and it is without value attribute.

size :- defines the size for specific input feild.

checked :- By default the checkbox will get selected without even selecting it. and it is without value attribute.

Why validation attribute are important ?

- Improve data accuracy.
- prevent wrong form sub mission
- Reduce Javascript validation code
- Improve user experience.